GOVERNMENT POLYTECHNIC DEEH SADAR

UNNAO

Diploma in Electrical & Electronics Engineering

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SYNOPSIS OF MINI PROJECT

BLUETOOTH BASED HOME THEATRE SYSTEM

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Introduction

The Bluetooth Based Home Theatre System is a wireless audio entertainment system designed to provide high-quality sound output with attractive visual effects. The system receives audio signals through a Bluetooth panel and amplifies them using a 3-IC amplifier board based on TDA2030 audio amplifier ICs. The amplified audio is reproduced through two satellite speakers and one subwoofer speaker.

To enhance the listening experience, LED strip lights are connected to the bass output section. The LED lights glow and vary in intensity according to the bass level of the music, creating an attractive lighting effect.

Objectives

- To design a wireless Bluetooth home theatre system.
- To provide clear and powerful audio output.
- To create bass effects using a dedicated subwoofer.
- To synchronize LED lighting with bass beats.
- To develop a low-cost entertainment system using readily available electronic components.

Components Required

S.No.	Component	Quantity
1	TDA2030 Based 3-IC Amplifier Board	1
2	12-0-12V Transformer (3A)	1
3	Bluetooth Audio Panel	1
4	3-inch Speaker	2
5	5.5-inch Subwoofer Speaker	1
6	Potentiometer (Volume Control)	1
7	LED Strip Light	1

8	Rainbow Wire	As Required
9	1.5 mm Wire	As Required
10	Three-Pin Top	1
11	Ply Board (Cabinet)	As Required
12	Lamination Paper	As Required
13	Insulation Tape	As Required

Block Diagram

AC Supply

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12-0-12V Transformer

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Power Supply Section

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Bluetooth Audio Panel

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TDA2030 Amplifier Board

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Up Speaker + Down Speaker + Subwoofer

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Audio Output

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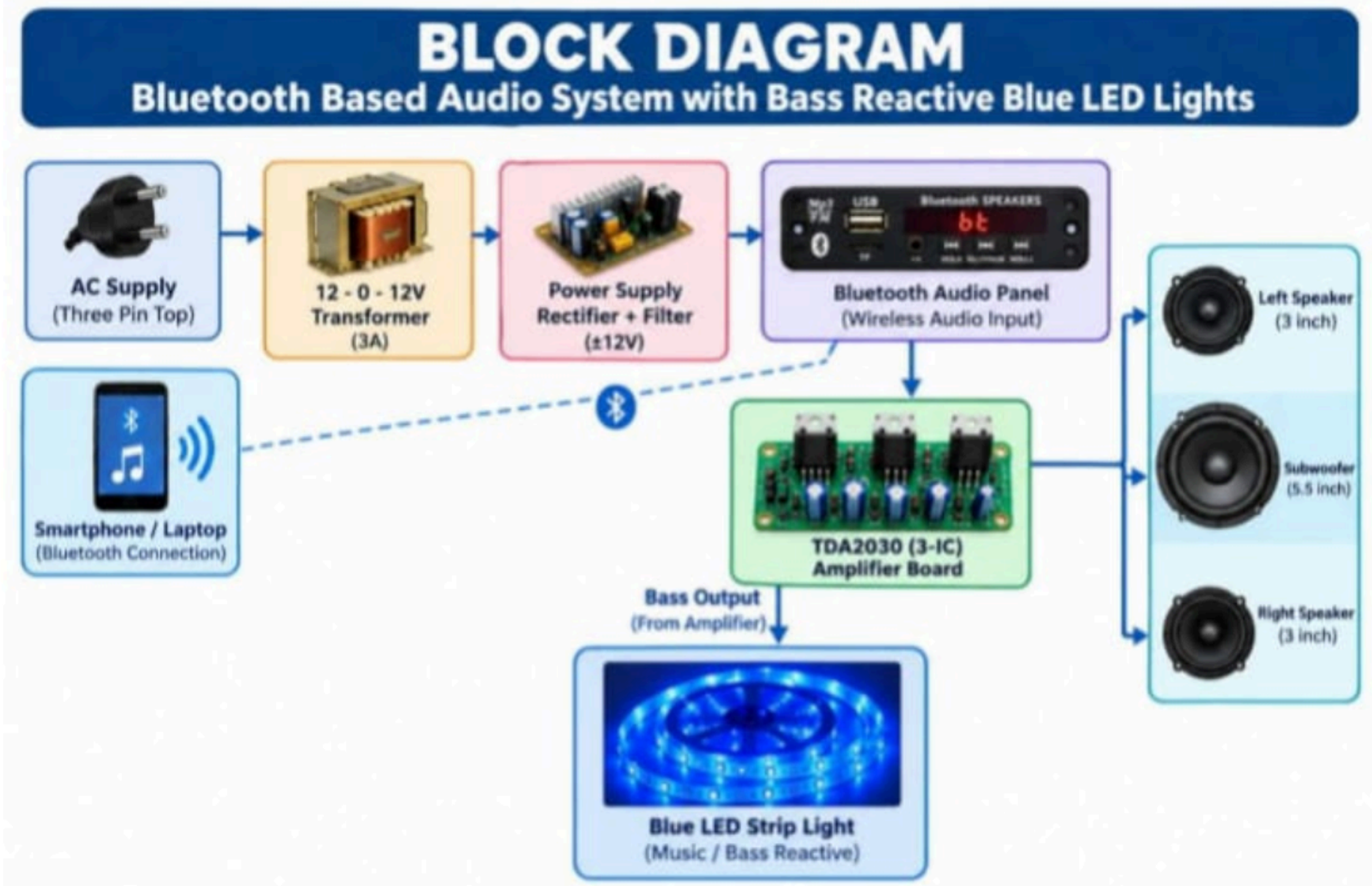
Bass Signal

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LED Strip Light

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Lighting Effect



Working Principle

The system operates on AC mains power supplied through a three-pin plug. The 12-0-12V, 3A transformer provides the required power to the amplifier board.

The Bluetooth panel receives audio signals wirelessly from smartphones, laptops, or tablets. The audio signal is sent to the TDA2030 amplifier board where it is amplified.

The amplified signal is distributed to:

- Two 3-inch speakers for left and right channel audio.
- One 5.5-inch subwoofer for low-frequency bass sounds.

The LED strip is connected to the bass output section. Whenever strong bass frequencies are produced, the LED strip glows brighter and flashes according to the music beats, creating an attractive visual effect.

Features

- Wireless Bluetooth connectivity.
- Stereo sound output.

- Dedicated subwoofer for deep bass.
- Bass-reactive LED lighting.
- Portable and compact design.
- Easy operation and maintenance.

Applications

- Home entertainment systems.
- Music playback systems.
- Small parties and events.
- Hostel rooms and study rooms.
- Multimedia audio systems.

Advantages

- No audio cables required.
- Good sound quality.
- Attractive bass lighting effects.
- Low construction cost.
- Easy Bluetooth pairing.
- User-friendly operation.

Future Scope

- Remote control operation.
- Rechargeable battery backup.
- RGB music-controlled lighting.
- Digital display panel.
- Wi-Fi audio streaming support.
- Surround sound enhancement.

Conclusion

The Bluetooth Based Home Theatre System with Bass Reactive LED Lights successfully combines wireless audio technology with visual lighting effects. The system provides quality sound through stereo speakers and a subwoofer while creating an enhanced entertainment experience using LED lights that respond to bass frequencies. The project is economical, practical, and suitable for educational as well as domestic applications.



FINAL MINI PROJECT